MZ2POL-02: Understanding the New Access Control Model

> **Series:** MZ2POL | **Notebook:** 3 of 8 | **Created:** December 2025 | **Last Updated:** 01/28/2026

Overview

This notebook provides a deep dive into the **ABAC (Attribute-Based Access Control)** framework that replaces Management Zones for access control. You'll learn how Policies, Boundaries, and Segments work together to provide flexible, scalable access management.

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Prerequisites

- Completed MZ2POL-01: Introduction
- Access to Dynatrace Account Management
- Understanding of current Management Zone configuration

Learning Objectives

By the end of this notebook, you will:
1. Understand the ABAC framework architecture
2. Know the relationship between Policies, Boundaries, and Segments
3. Understand how permissions flow through the system
4. Be able to map MZ concepts to the new model

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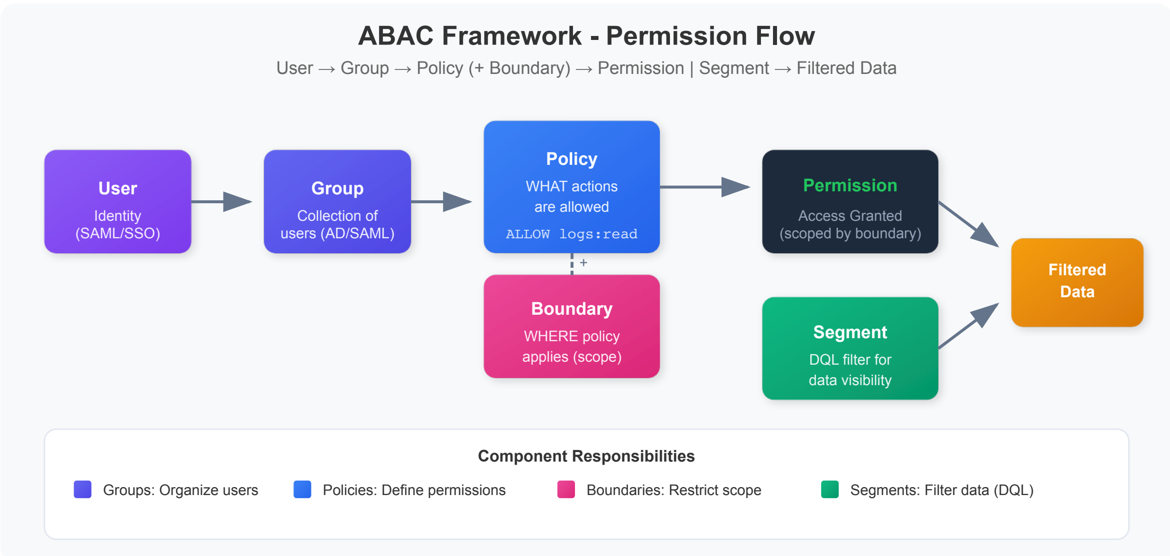
1. ABAC Framework Architecture

The Permission Flow



Key Components

Component	Purpose	Configured In
Users	Individual identities	Account Management
Groups	Collections of users	Identity & Access Management
Policies	Permission definitions	Policy Management
Boundaries	Scope restrictions	Policy Boundaries
Segments	Data filtering	Segments app

How It Works Together

1. **Users** are assigned to **Groups**
2. **Groups** are bound to **Policies**
3. **Boundaries** can optionally restrict the **Policy** scope
4. **Segments** filter what data users see (independent of permissions)

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2. Policies Deep Dive

What Are Policies?

Policies are the core of ABAC - they define **WHAT** users can do.

Policy Types

Type	Description	Editable
Default Policies	Pre-defined by Dynatrace	No (read-only)
Custom Policies	Created by administrators	Yes

Default Policies Categories

Dynatrace Access Policies (Platform features):
- `Dynatrace Viewer` - Read-only access
- `Dynatrace Standard User` - Standard operations
- `Dynatrace Professional User` - Advanced features
- `Dynatrace Admin User` - Full administration

Data Access Policies (Monitored data):
- `Data Viewer` - Read monitored data
- `Data Editor` - Modify data configurations

Policy Statement Structure

```
ALLOW &lt;service&gt;;:&lt;resource&gt;;:&lt;action&gt;; [WHERE
&lt;condition&gt;;]
```

Examples:

```
ALLOW storage:buckets:read
ALLOW settings:objects:read WHERE settings:schemaId =
"builtin:alerting.profile"
ALLOW storage:logs:read WHERE storage:dt.security_context = "team-a"
```

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3. Boundaries Deep Dive

What Are Boundaries?

Boundaries restrict **WHERE** policies apply - they limit the scope of permissions.

Key Characteristics

- **Optional** but powerful for fine-grained access control
- Work **together** with policies (not standalone)
- **Further restrict** existing policy permissions
- Enable **reusability** across multiple policy assignments

Boundary Query Syntax

```
&lt;field&gt; &lt;operator&gt; "&lt;value&gt;"
```

Supported Operators:
- `=` - Equals
- `!=` - Not equals
- `startsWith` - Prefix match
- `in` - Value in list

Common Fields:
- `environment` - Environment restrictions
- `environment:management-zone` - MZ-based restrictions
- `storage:dt.security_context` - Security context filtering
- `storage:bucket.name` - Bucket-based restrictions

Boundary Examples

Restrict to specific Management Zone (transitional):

```
environment:management-zone = "Production-NA"
```

Restrict by Management Zone prefix:

```
environment:management-zone startsWith "mgmt_na"
```

Restrict by Security Context:

```
storage:dt.security_context = "team-frontend"
```

Restrict by Bucket (data partitioning):

```
storage:bucket.name IN ("frontend_logs", "frontend_spans")
```

Boundary Limitations

Limitation	Workaround
Max 10 restrictions per boundary	Create multiple boundaries
No AND operator (lines are OR)	Use multiple boundary assignments
Only works with security policies	Cannot use with role-based permissions

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3.5 Buckets: Data Partitioning for Access Control

What Are Buckets?

Buckets are Grail storage containers that physically partition your data. Unlike boundaries (which filter at query time), buckets provide **hard data separation** - data in one bucket is completely isolated from other buckets.

Why Use Buckets for Access Control?

Benefit	Description
Physical isolation	Data is stored separately, not just filtered
Performance	Smaller, team-specific data sets query faster
Compliance	Meet regulatory requirements for data separation
Retention control	Different retention policies per bucket
Cost allocation	Track storage costs by team/project

Buckets vs Boundaries vs Segments

Aspect	Buckets	Boundaries	Segments
Isolation	Physical separation	Query-time filtering	Query-time filtering
Reversible	❌ No (data can't move)	✅ Yes	✅ Yes
Performance	Best (smaller data sets)	Good	Good
Flexibility	Low (plan carefully)	High	High
Use case	Team data isolation	Permission scoping	Data filtering

Bucket Limitations

> ⚠️ **Critical:** Plan buckets carefully - these constraints cannot be changed later.

Limitation	Details
One data type per bucket	Logs, metrics, events, OR spans - not mixed
Names are immutable	Bucket names CANNOT be changed after creation
No data migration	Data CANNOT be moved between buckets
Naming rules	3-100 chars, lowercase alphanumeric, underscores, hyphens only
Maximum buckets	80 per environment (default limit)
Optimal ingest	~1 TB/day per bucket for best query performance
Acceptable ingest	1-3 TB/day per bucket (limited query window)
Maximum ingest	3 TB/day per bucket hard limit

Query Constraints

Limit	Value	Impact
Maximum data scanned	500 GB	Limits queryable time window
Maximum records returned	1,000	Use aggregations for larger datasets
Maximum response payload	1 MB	Large result sets may be truncated

Using Buckets in Policies

Grant team access to their specific buckets:

```
// Policy: Grant team access to their buckets
ALLOW storage:logs:read WHERE storage:bucket.name = "frontend_logs"
ALLOW storage:spans:read WHERE storage:bucket.name = "frontend_spans"
ALLOW storage:metrics:read WHERE storage:bucket.name = "frontend_metrics"
```

Using Buckets in Boundaries

Combine buckets with security context for layered control:

```
// Boundary: Restrict to team's buckets AND security context
storage:bucket.name IN ("frontend_logs", "frontend_spans",
"frontend_metrics");
storage:dt.security_context IN ("team-frontend");
environment:management-zone IN ("Frontend-Team");
```

When to Use Buckets vs Other Options

Scenario	Recommended Approach
Strict team data isolation	**Buckets** + Policies
Compliance/regulatory requirements	**Buckets** for physical separation
Flexible team access	**Boundaries** with security context
Ad-hoc data filtering	**Segments**
Combination (defense in depth)	**Buckets** + **Boundaries**

MZ to Bucket Mapping

MZ Pattern	Bucket Strategy
Team MZs	Team-specific buckets: `teamA_logs` , `teamA_spans`
Environment MZs	Environment buckets: `prod_logs` , `dev_logs`
Regional MZs	Region buckets: `useast_logs` , `euwest_logs`
Compliance MZs	Compliance buckets: `pci_logs` , `hipaa_logs`

> **For comprehensive bucket guidance**, see **ORGNZ-03: Bucket Strategy and Design**.

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4. Segments Deep Dive

What Are Segments?

Segments are **DQL-based filter conditions** that control what data users see - they're the replacement for MZ data filtering.

Key Characteristics

- **Query-time evaluation** (not precalculated)
- **Multi-dimensional** - can layer multiple segments
- **DQL-powered** - full query language flexibility
- Support **variables** for dynamic filtering
- **Independent of permissions** - filtering only

How Segments Work in DQL

When a segment is applied, Grail:
1. Evaluates segment conditions relevant to the query
2. Applies filters based on the targeted data object
3. Multiple conditions for same data object = OR combined

Segment vs. Management Zone Filtering

Aspect	Management Zone	Segment
Evaluation	Precalculated	Query-time
Performance	Bottleneck at scale	Highly scalable
Flexibility	Fixed rules	Dynamic DQL
Variables	No	Yes
Multi-dimensional	No	Yes

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5. Querying Current Access Configuration

View Services with Security Context

Security Context is key for access control in the new model:

```
// List services and their security context
// Security context is used for fine-grained access control
fetch dt.entity.service
| fields entity.name,
         dt.security_context,
         managementZones
| filter isNotNull(dt.security_context)
| sort entity.name asc
| limit 50
```

Analyze Entity Types and Their Attributes

Understanding entity attributes helps design effective segments:

```
// Analyze host entity attributes for segment planning
// Tags and metadata are useful for segment conditions
fetch dt.entity.host
| fields entity.name,
         tags,
         managementZones
| limit 20
```

Check Kubernetes Cluster Distribution

K8s clusters often map to organizational boundaries:

```
// List Kubernetes clusters with their attributes
// Clusters often align with team or environment boundaries
fetch dt.entity.kubernetes_cluster
| fields entity.name,
         tags,
         managementZones
| sort entity.name asc
```

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6. Mapping MZ Concepts to New Model

Common MZ Patterns and Their Replacements

MZ Pattern	New Approach
Team-based MZs	Security Context + Policies
Environment MZs (Dev/Prod)	Boundaries with environment filters
Region MZs	Segments with cloud region filters
Application MZs	Segments with service/app filters
Multi-tenant MZs	Boundaries + Security Context

Example: Team-Based Access Control

Old (Management Zone):
- MZ: "Team-Frontend" with rules for frontend services
- Users assigned to MZ get filtered view

New (Policies + Boundaries + Segments):
1. **Policy**: `Dynatrace Standard User` or custom policy
2. **Boundary**: `storage:dt.security_context = "team-frontend"`
3. **Segment**: DQL filter for frontend services

Example: Environment Separation

Old (Management Zone):
- MZ: "Production" with host/service rules
- MZ: "Development" with different rules

New (Policies + Boundaries + Segments):
1. **Policy**: Same policy for both groups
2. **Boundary (Prod)**: Environment-specific restrictions
3. **Boundary (Dev)**: Environment-specific restrictions
4. **Segments**: Environment-based data filters

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7. Access Control Decision Flow

Permission Evaluation Order

```
1. User attempts action
2. System checks user's group memberships
3. For each group, evaluate bound policies
4. Apply boundary restrictions (if any)
5. If ALLOW found with matching conditions → Permit
6. If no ALLOW found → Deny (implicit)
```

Segment Application

```
1. User queries data (DQL, dashboard, app)
2. Active segment(s) identified
3. Segment conditions injected into query
4. Grail evaluates with segment filters
5. Filtered results returned
```

Key Differences from MZ

MZ Behavior	New Behavior
Single construct for access + filtering	Separate concerns (Policies vs Segments)
Precalculated membership	Runtime evaluation
Limited to entity types	Any DQL-queryable attribute
Flat structure	Hierarchical (groups → policies → boundaries)

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8. Best Practices for the New Model

Policy Design

1. **Start with default policies** - customize only when needed
2. **Use least privilege** - grant minimum required permissions
3. **Group similar permissions** - avoid policy sprawl
4. **Document policy purpose** - maintain clarity

Boundary Design

1. **Create reusable boundaries** - one boundary, many uses
2. **Use meaningful names** - indicate scope clearly
3. **Keep conditions simple** - easier to audit
4. **Leverage security context** - for entity-level control

Segment Design

1. **Align with business structure** - teams, regions, products
2. **Use variables** - for dynamic, flexible filters
3. **Test thoroughly** - verify filtering works as expected
4. **Layer segments** - combine for precise filtering

Summary

In this notebook, you learned:

1. **ABAC Framework**: How Users → Groups → Policies → Permissions flow
2. **Policies**: Define WHAT users can do (permissions)
3. **Boundaries**: Restrict WHERE policies apply (scope)
4. **Segments**: Filter WHAT data users see (DQL-based)
5. **Mapping**: How MZ patterns translate to the new model

Next Steps

Continue to **MZ2POL-03: Assessment and Migration Planning** to:
- Audit your current MZ configuration in detail
- Create a migration mapping document
- Plan the phased migration approach

Additional Resources

- Working with Policies
- IAM Policy Reference
- Default Policies Reference
- Grant Access to Entities with Security Context

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